TaSain Thomas

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SUMMARY

AI/ML Engineer bridging published research and production systems. Translated peer-reviewed virtual-anthropometry literature into a secure VLM evaluation pipeline, extending pose-to-ratio methodology with GPT-as-Judge reasoning and LoRA fine-tuning - achieving 89% BMI estimation accuracy at 95% training cost reduction. Industry 4.0 automation background delivering 33% throughput gains at a major automotive manufacturer. Seeking to apply ML validation, research translation, and adversarial robustness expertise to safety-critical AI systems.

EDUCATION

Wayne State University, James and Patricia Anderson College of Engineering **May 2026**
Bachelor of Science, Computer Science *Detroit, MI*
- **GPA:** 3.4/4.0

CORE COMPETENCIES

CORE COMPETENCIES: ML Validation, CI/CD, Red Teaming, Model Monitoring, LoRA/VLM Fine-Tuning, RAG--
, Simulation-Based Evaluation, Uncertainty Quantification, Kubernetes, Docker, Python, PyTorch, C++, Adversarial Robustness
TECHNICAL SKILLS: Python, C++, JavaScript, SQL, Java, Dart, PyTorch, TensorFlow, Hugging Face, LangChain-, LoRA/QLoRA, RAG, Vector Databases, Transformer Architectures, Kubernetes, Docker, CI/CD, Model Serving & Monitoring, Helm, Terraform, GPU Passthrough, AWS/Azure, DigitalOcean, Nginx, Cloudflare, Proxmox, systemd, Red Teaming, Prompt Injection Defense, Zero Trust, Adversarial Robustness, Kali Linux

PROFESSIONAL EXPERIENCE

Magna EV Structures **Sep 2023 - Sep 2024**
Industry 4.0 / Automation Intern (Rotational) *St. Clair, MI*
- Delivered four automation projects on schedule by integrating real-time production data pipelines via custom API development.
- Improved throughput 33% by implementing Vuforia Model Targets for vision-guided assembly verification.
- Built C++/SQLite3 worker-to-station matching engine with evaluateFit() scoring algorithm, reducing assignment time by 40%.
- Reduced debug cycle time 25% by developing custom C# validation logic that automated error handling and root-cause identification.

ENGINEERING PROJECTS

ConformalBMI: Uncertainty-Aware Body Mass Estimation from Images GitHub **May 2026 - Aug 2026**
- Translated published computer-vision research (DigitalScale, ETH Zurich) into a functional BMI-estimation prototype, adapting a Squeeze-and-Excitation DenseNet (SE-DenseNet121) in PyTorch.
- Trained on Celeb-FBI (7,208 real images, 6,174 labeled); final performance of 2.81 BMI-point MAE and 11.56% MAPE on held-out test data.
- Extended the model with split conformal prediction for distribution-free uncertainty -- 90% prediction intervals (q = 6.44) with empirically verified 90.4 coverage on unseen samples.
- Layered an LLM-as-judge reasoning module (GPT-4.1-Nano) to flag high-uncertainty and extreme-BMI tail cases, converting a point estimate into a calibrated, explainable prediction.

MLOps Infrastructure & AI Agent Engineering **Mar 2026 - Present**
- Architected auto-scaling Kubernetes inference cluster with GPU passthrough serving LLMs at 99.9% uptime, with model monitoring and real-time incident response.
- Embedded red-team security testing (Kali Linux) into CI/CD, patching prompt injection vulnerabilities pre-production.

SaaS Booking & Payment Platform GitHub **Mar 2026 - Present**
- Engineered full-stack CRM (Python/JavaScript) with Stripe payments and Telegram Bot API; deployed on DigitalOcean, Nginx, systemd, Cloudflare CI/CD at 24/7 uptime.

Real-Time Mobile Collaboration Platform **Jan 2026 - May 2026**
- Developed cross-platform (Flutter/Dart) collaborative workspace integrating ZegoCloud real-time video and RevenueCat subscription monetization; shipped to production for client deployment.